to support multiple processors there is a network infrastructure to support multiple computer units. The programs running on the different computers can communicate by passing messages using the network.

An alternative type of multicomputer system is cluster computing using PCs (sometimes referred to as a 'server farm').

Whatever the technology used, these systems have an extremely large number of individual processors working in parallel.

Extension Question 18.02

The IBM Sequoia and a Beowulf cluster are different examples of cluster computing. You might wish to carry out some research to get some details of their computing power.

18.04 Virtual machines

A virtual machine is software not hardware. The most usual type of virtual machine is the **system virtual machine** which is software that emulates the hardware of a real computer system.

KEY TERM

System virtual machine: the emulation of computer system hardware using software

Remember that when a virtual machine is not being used an application program requires support from an operating system in order for the program to run on the hardware. The principle of a virtual machine is that a process interacts directly with a software interface provided by an operating system. The logical structure for the operation of a system virtual machine is shown in Figure 18.05.

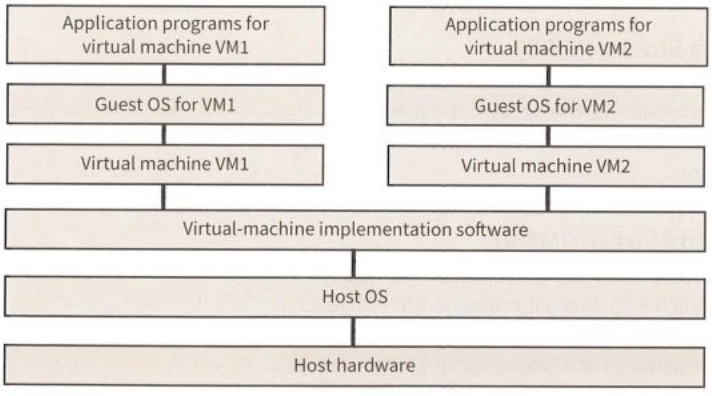


Figure 18.05 Logical structure for a system virtual machine implementation

The following points should be noted when looking at Figure 18.05.

1 The application programs are installed with the assistance of a guest OS. This guest OS will support the running application by interacting with the virtual machine as though it were the hardware that the guest OS would normally run on.

2 The virtual machine implementation software can be considered to be a utility program which, when running, is supported by the particular host OS which is specific to the host hardware.

3 There can be application programs running at the same time directly on the host hardware under the control of the host OS.

The main advantage of the virtual machine approach is that more than one different operating system can be made available on one computer system. This is particularly valuable if an organisation has legacy systems and wishes to continue to use the old software but does not wish to keep the old hardware. Alternatively, the same operating system can be made available many times by companies with large mainframe computers that offer server consolidation facilities. Different companies can be offered their own virtual machine running as a server.

One drawback to using a virtual machine is the time and effort required for implementation. Another drawback is the fact that the implementation will not offer the same level of performance that would be obtained on a normal system.

Discussion Point:

Are there other advantages or disadvantages in using a system virtual machine?

The Java virtual machine discussed in Chapter 8 (Section 8.05) is an example of a process virtual machine, based on a different underlying concept. The process virtual machine provides a platform-independent programming environment that allows a program to execute in the same way on any platform. This is specific software that only supports running a Java program. A system virtual machine supports any application.

Reflection Point:

The description 'virtual' occurs in several places in this book. Are you clear about how these different examples have their own specific context and meaning?

Summary

- A control unit can be hard-wired or microprogrammed.
- RISC (Reduced Instruction Set Computers) processors have a number of advantages compared to CISC (Complex Instruction Set Computers).
- Pipelining is one of the reasons for choosing a RISC architecture.
- Parallelism can be based at the instruction level, processor level or computer level.
- A system virtual machine is software emulating hardware.
- The Java Virtual Machine is an example of a process virtual machine.